PURITY IS PRESERVED BY A QUASI-FINITE BASE CHANGE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Quasi-finiteness is preserved by composition, and the characterization of purity by impurities. Morphisms, label morphisms-lemma-composition-quasi-finite (proof line 4921); Same chapter flat.tex, definition labelled definition-pure and the characterization following it (proof line 4924). The first shows that an impurity over the base change with the composite morphism quasi-finite gives an impurity over the base, and the characterization turns the non existence of impurities into purity.

Source reference. Original definition 05J4: sources/stacks-flat/pinned-original-source.tex, source lines 4808–4808. Navigation pointer for \ref{definition-pure} used inside the counted span (line 4924); the argument that consumes it is retained in the span.

Source reference. Original lemma 01TL: sources/stacks-chow/tags.txt, source lines 2378–2378. Navigation pointer for \ref{morphisms-lemma-composition-quasi-finite} used inside the counted span (line 4921); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes which is of finite type. Let $\mathcal{F}$ be a finite type quasi-coherent $\mathcal{O}_X$-module. Let $s \in S$. Let $(S', s') \to (S, s)$ be a morphism of pointed schemes. If $S' \to S$ is quasi-finite at s' and $\mathcal{F}$ is pure along X_s, then $\mathcal{F}_{S'}$ is pure along $X_{s'}$.*

Proof. It $(T \to S', t' \rightsquigarrow t, \xi)$ is an impurity of $\mathcal{F}_{S'}$ above s' with $T \to S'$ quasi-finite at t, then $(T \to S, t' \to t, \xi)$ is an impurity of $\mathcal{F}$ above s with $T \to S$ quasi-finite at t, see Morphisms, Lemma 01TL. Hence the lemma follows immediately from the characterization (2) of purity given following Definition 05J4. □

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